

PRATEEK KAPOOR

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SUMMARY

Backend engineer with 3+ years building Python microservices, data pipelines, and cloud-native systems. Experienced with async workflows, AWS/Azure services, and applied AI integrations including RAG pipelines, LLM-based document processing, and ML model inference. Recently focused on automating complex document workflows using Azure Functions, Document Intelligence, and open-source ML models.

EXPERIENCE

Sr. Full Stack Developer

July 2025 – Present

Simplicity Group | Summit, NJ

- Built AI-powered PDF form field detection on Azure Functions using a YOLOv11-based ML model with fan-out/fan-in parallelism to process 50+ forms simultaneously, achieving 95% field detection accuracy
- Developed an LLM-based field naming pipeline using Azure Document Intelligence for text layout extraction and a custom interleaving algorithm to contextualize form fields, reducing carrier onboarding from 2+ weeks to 2 days for a single BA
- Architected FastAPI Python microservice for PDF form validation deployed on Azure Functions, using async batch workflows with blob storage and queue processing to automate detection of mapping defects across 50+ form variants, targeting 20% of insurance carrier onboarding tickets
- Developed automated ETL pipeline syncing 300+ insurance form fields from third-party API with differential updates, transactional rollback, and idempotent logic, reducing 100+ hours of manual entry

Software Engineer

January 2023 – July 2025

Yale University | New Haven, CT

- Built a Python RAG ingestion pipeline (Azure OpenAI + AI Search) processing corpora ranging from textbooks to 3M-row Excel logs, proving feasibility and securing investment in a production chatbot platform
- Refactored legacy Ruby DNS registration to Go-based REST microservice, containerized and deployed to Amazon ECS, eliminating SOAP dependency

EDUCATION

M.S. in Computer Science

Expected: June 2027

Georgia Institute of Technology

B.S.E. in Computer Science and Engineering

2022

University of Connecticut | Storrs, CT

Major GPA: 3.7/4

Honors: Fall 2021 Dean's List, 2020 & 2021 New England Scholar

SKILLS

Programming Languages: Python, Go, Typescript

Frameworks: FastAPI, LangGraph, LangChain

Tools: Jupyter Notebooks, Docker, GitHub Actions, Git, Jenkins, Linux

Cloud: Azure (Foundry, Document Intelligence, Durable Functions, Storage), AWS (ECS, S3, EC2)